

innovate

achieve

lead



BITS Pilani
Pilani Campus

CE F230

Civil Engineering Materials

Concrete Workability



Workability of Concrete

- **Effort** required to manipulate a concrete mixture with a minimum segregation.
- Poor workability vs good workability
 - <http://youtu.be/tqP1DDYt2Gk>
 - <http://youtu.be/C9g5fAkpWNA>

Two Main Components of Workability



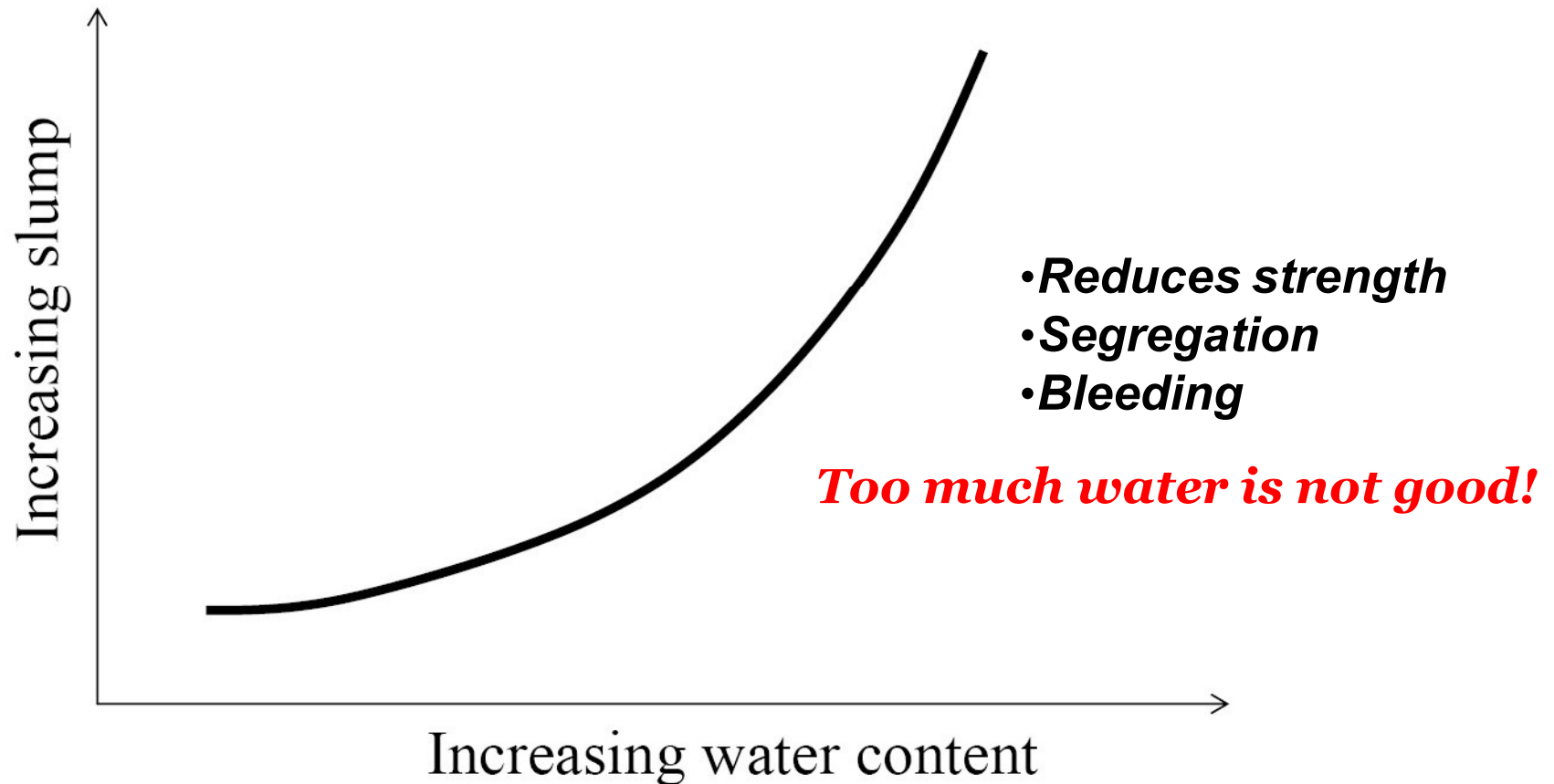
- **Consistency** describes the ease of flow.
 - **ACI.** The relative mobility or ability of freshly mixed concrete or mortar to flow
- **Cohesiveness or stability** describes the tendency to bleed or segregate.



Factors Affecting Workability

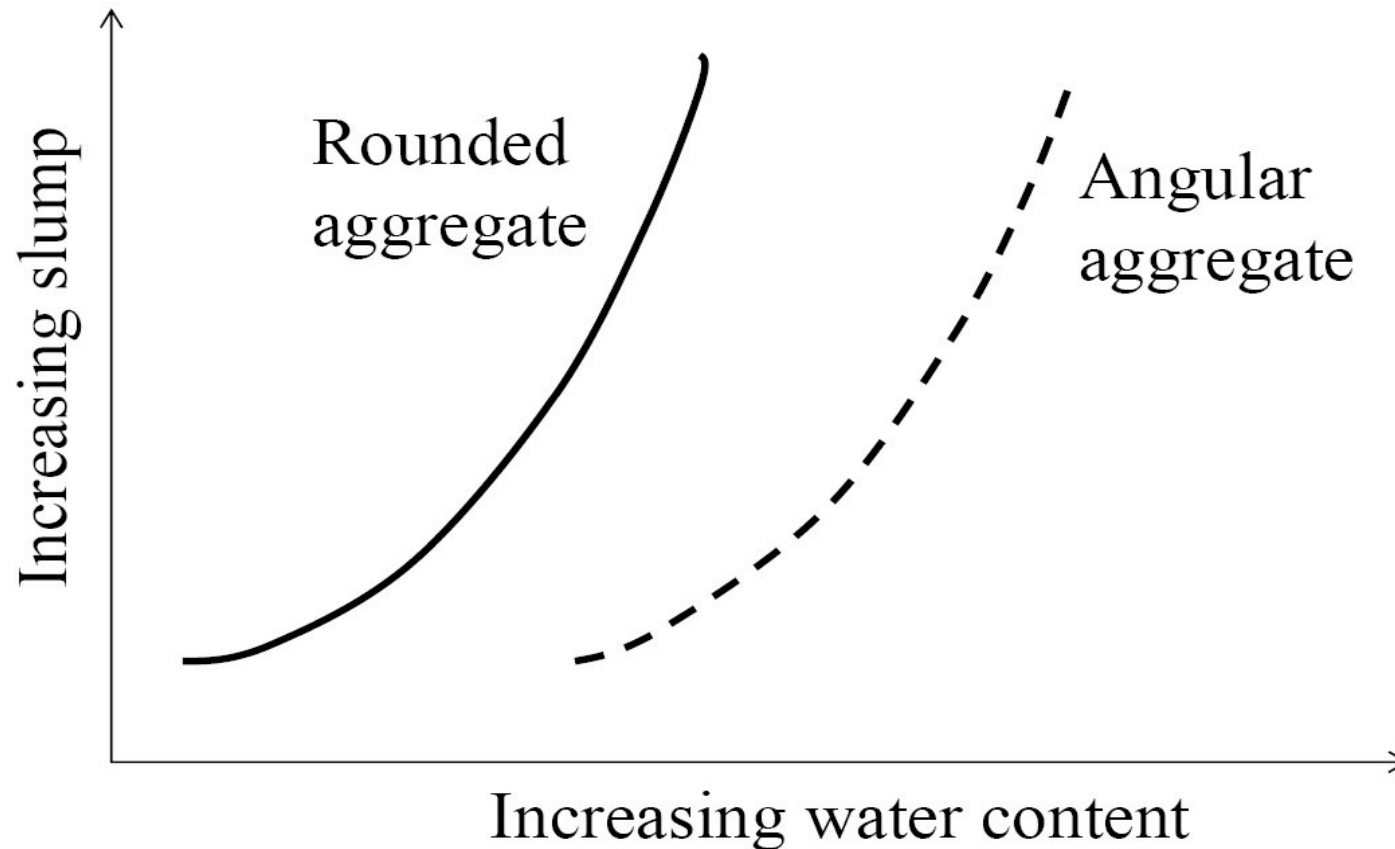
- ❖ Water content
- ❖ Aggregate type and grading
- ❖ Aggregate/cement ratio
- ❖ Presence of admixtures
- ❖ Fineness of cement
- ❖ Time
- ❖ Temperature

Effect of Water Content on Slump





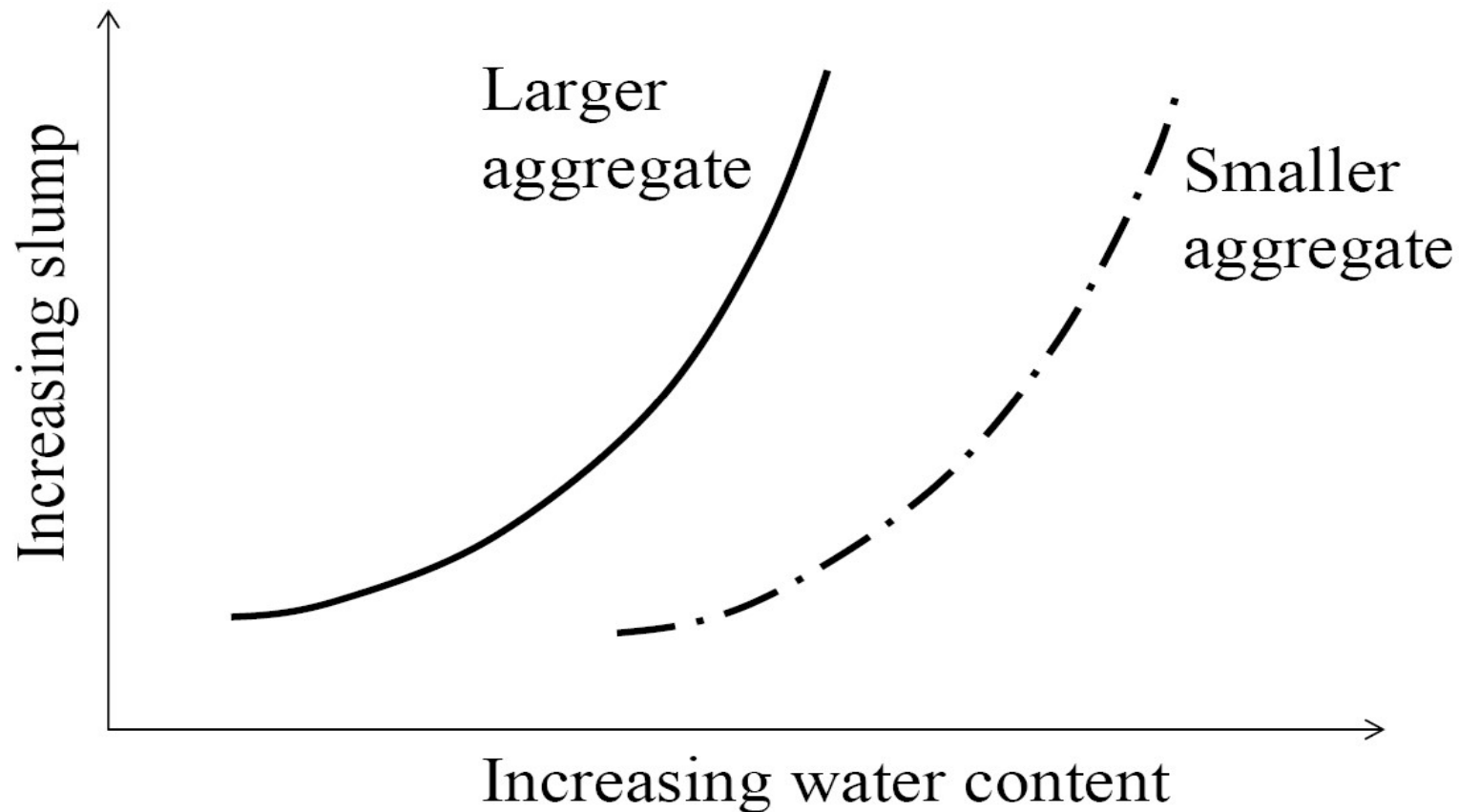
Effect of Agg. Shape on Slump



Roundness and smoothness of aggregates increases slump.



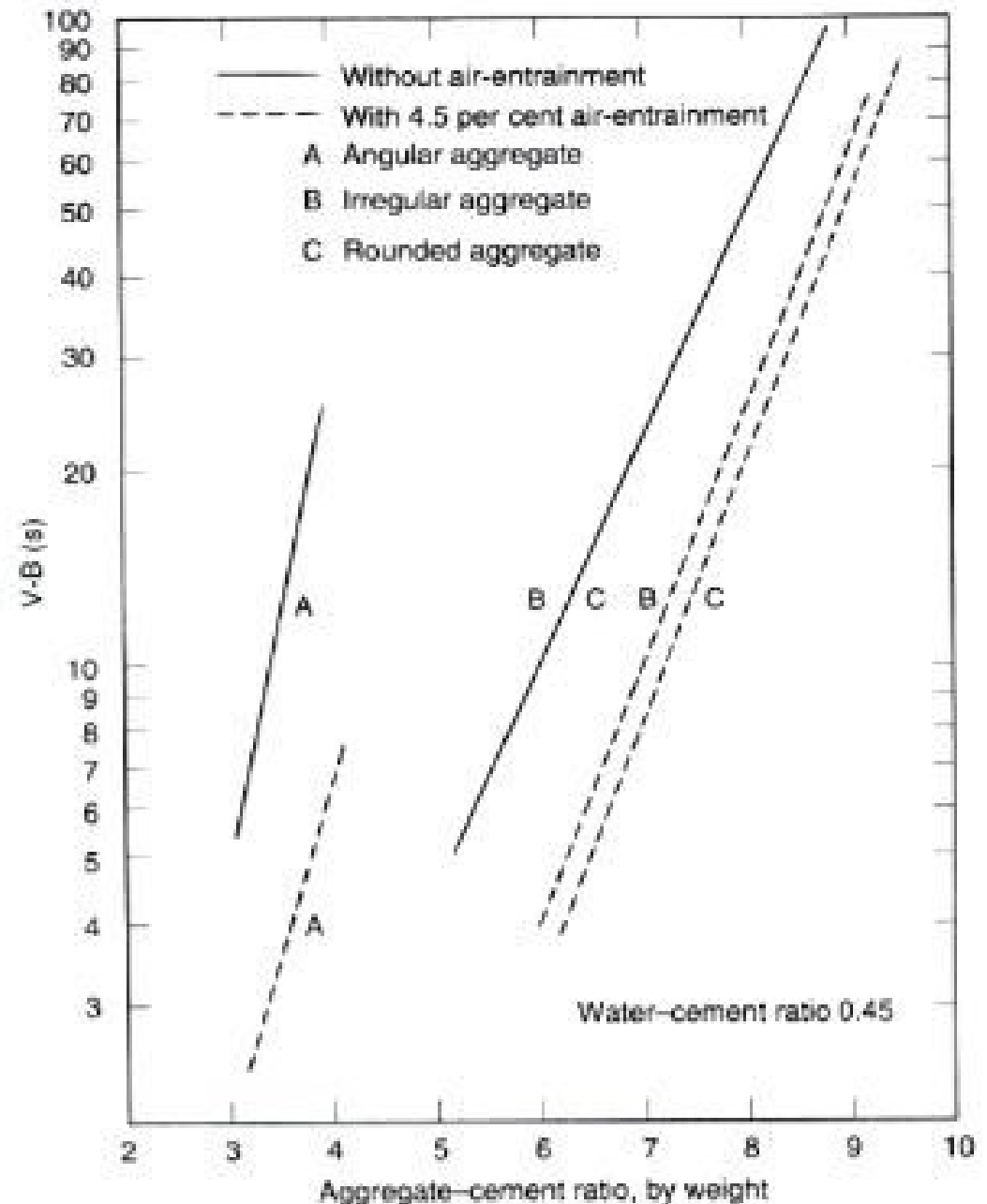
Effect of Agg. Size on Slump



Slump decreases as specific surface area of aggregate increases (smaller size), since this requires a greater proportion of water to wet aggregate particles, thus leaving a smaller amount of water for lubrication.

Influence of Aggregate to Cement Ratio on Workability

- For a constant w/c ratio, an increase in the agg/c ratio will decrease the workability
- Higher Vebe time (V-B) corresponds to lower workability

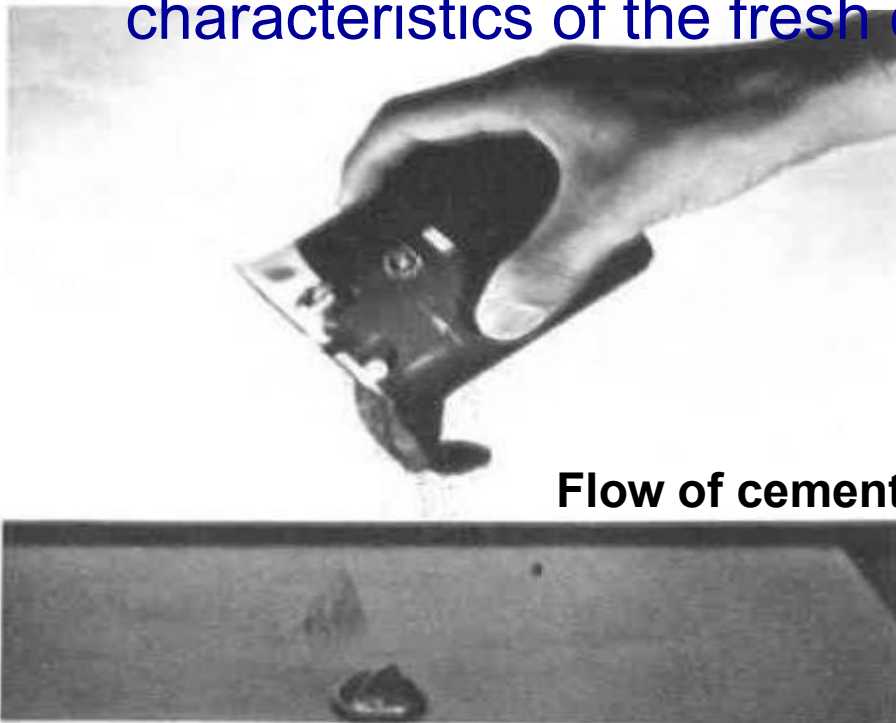


Source: Jackson & Dhir

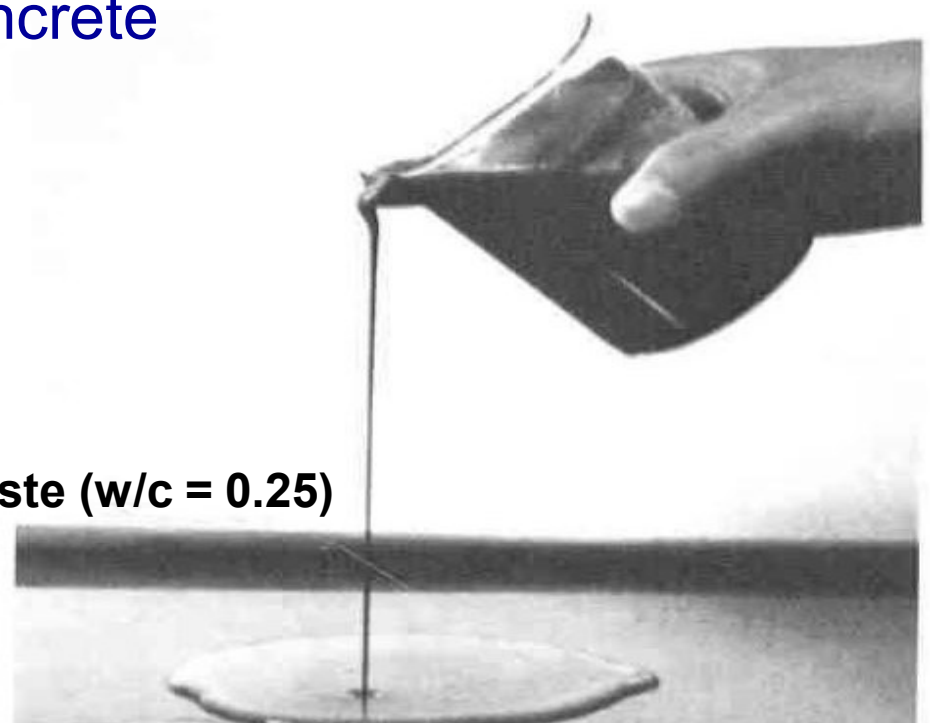
Effect of Admixtures on Workability



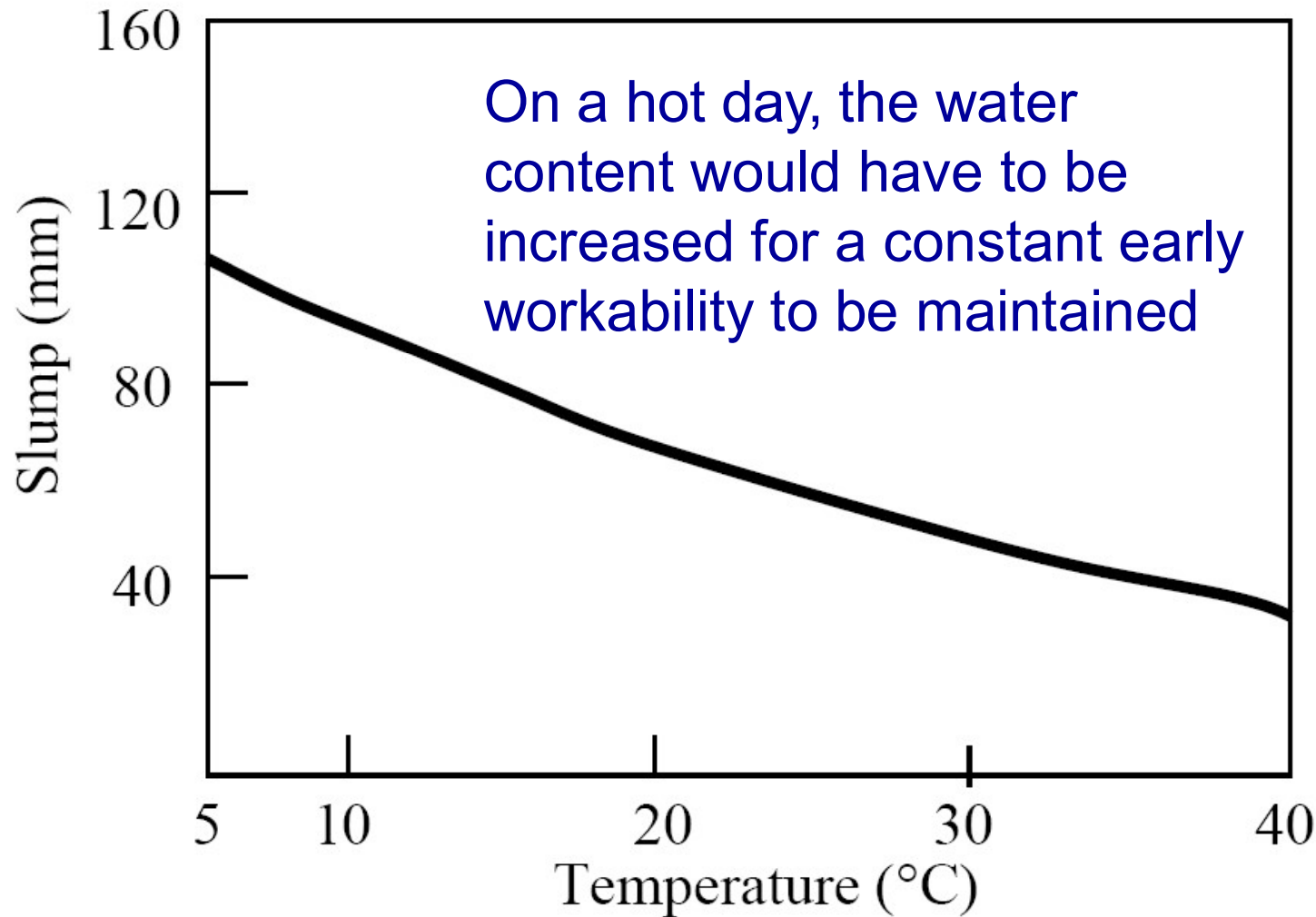
- ❖ The principal admixtures affecting improvement in workability of concrete are **water-reducing** and **air-entraining** agents as well as fly ash.
- ❖ The extent of increase in workability is dependent on the type and amount of admixture used and the general characteristics of the fresh concrete



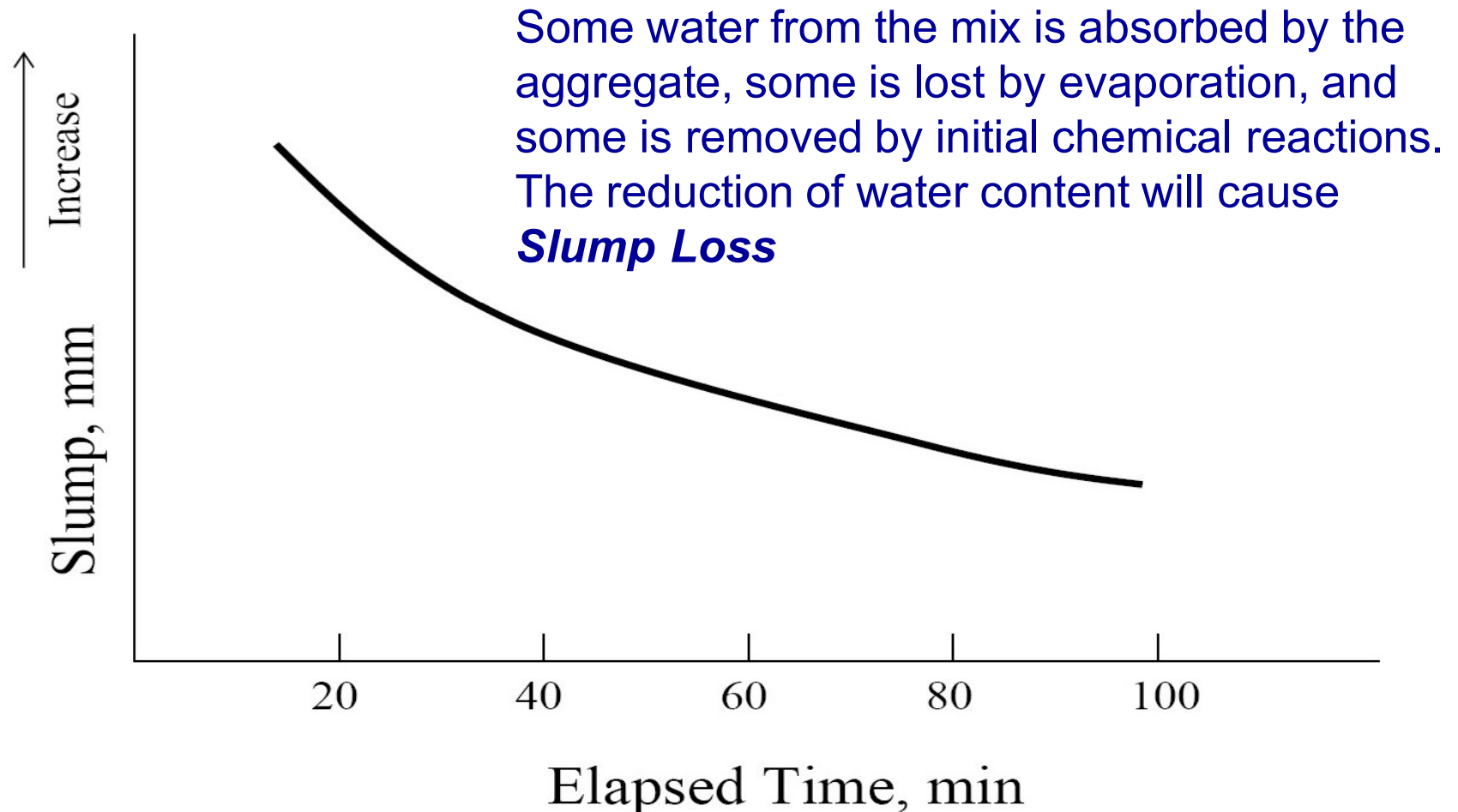
Flow of cement paste (w/c = 0.25)



Effect of Temperature on Workability



Effect of Time on Workability (Slump Loss)



Cohesion and Segregation

- Concrete with good workability ought to be **cohesive**, should not segregate
- **Segregation** can be defined as separation of constituents of a heterogeneous mixture so that their distribution is no longer uniform. Tendency for
 - Sand-cement mortar to separate from coarse aggregate
 - Cement paste to separate from fine aggregate

<http://youtu.be/aHNxq-Eda0c>

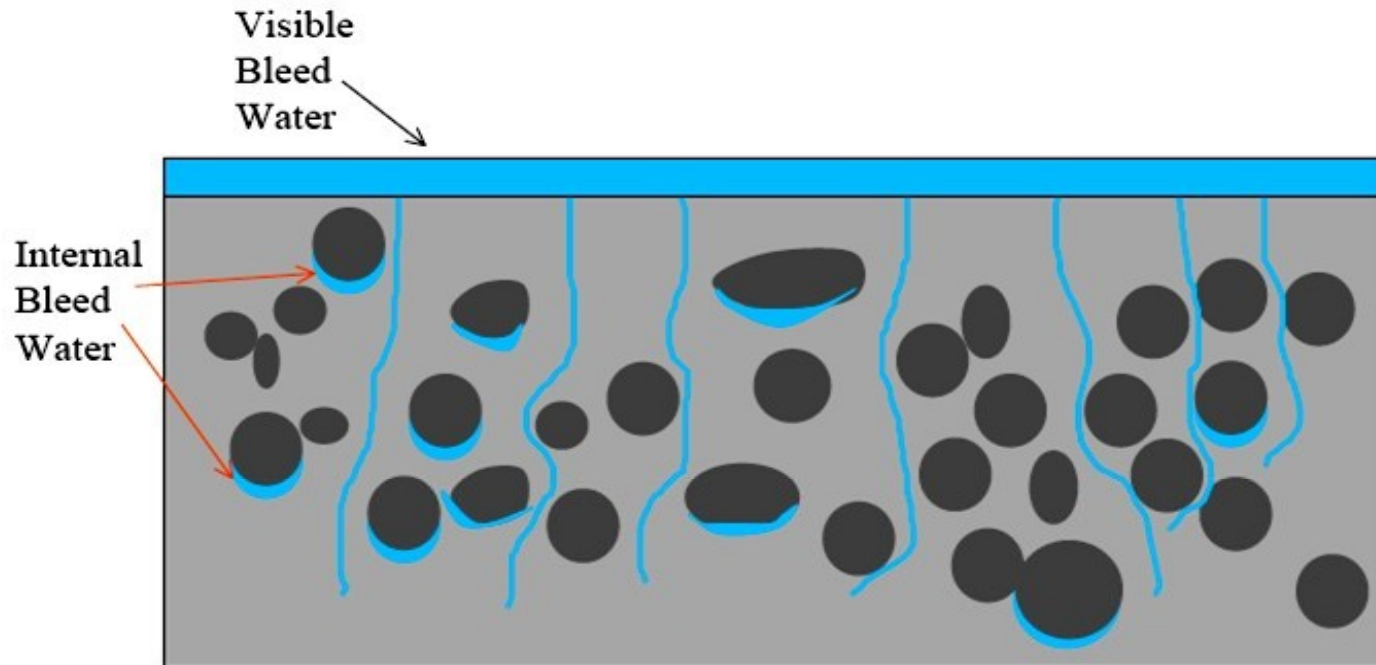




Bleeding (Water Gain)

- A special form of segregation in which some of the water in mix tends to rise to the surface of freshly placed concrete
- Create a porous and weak layer of non-durable concrete and zones of poor bond between cement paste and large aggregate particles or reinforcement

Bleeding



■ Caused by

- Lack of fines (300 μm and below)
- High free water content
- Water reducing admixture overdose

■ Control of bleeding

- Increasing cement fineness or using SCM (i.e. pozzolans)
- Increasing hydration rate (C_3A)
- Reducing free water content

Workability Tests

- ❖ Slump test
- ❖ Compacting factor test
- ❖ Vebe test



Slump Test (ASTM C 143): Principle



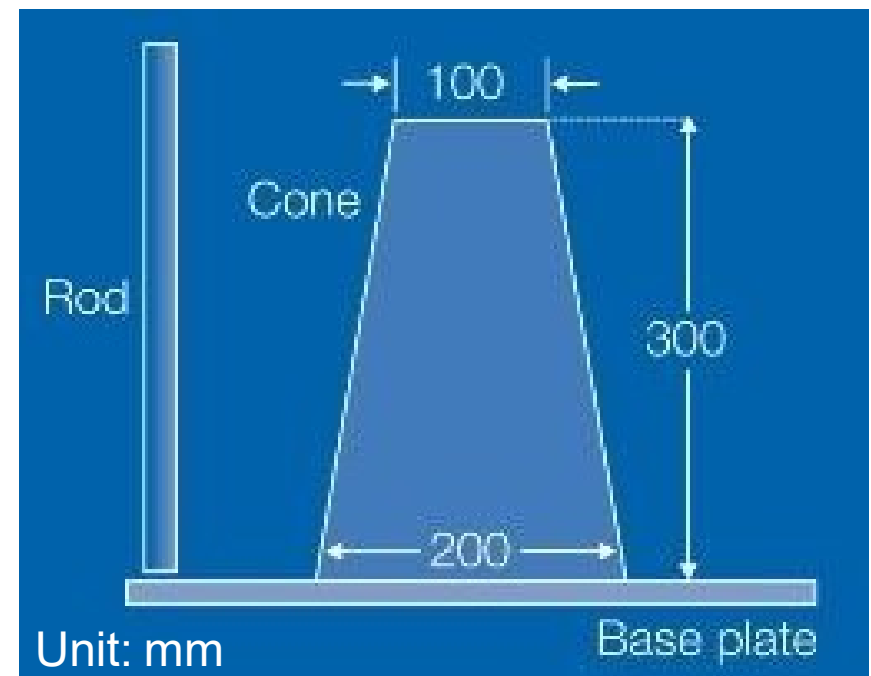
- The slump test is a measure of the behaviour of a compacted inverted cone of concrete under the action of **gravity**. It measures the consistency or the wetness of concrete.

Slump Test: Apparatus and Procedure



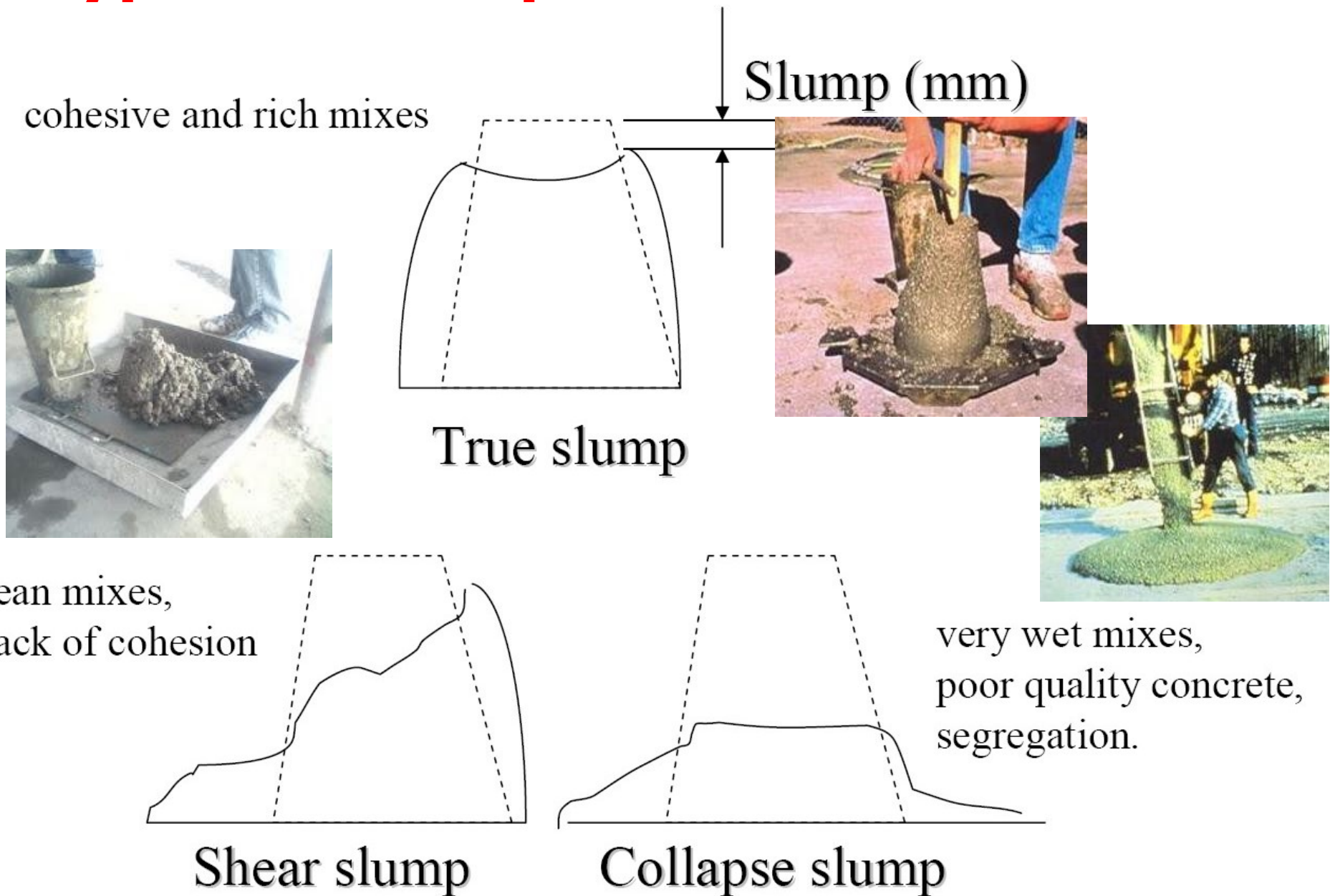
- Use inverted cone
- Fill it up with three layers of equal volume
- Rod each layer 25 times
- Scrape off surface
- Cone lift away vertically
- Slump measurement: Downward movement of the concrete

<http://youtu.be/Hmo7tMsRD1g>



Developed in US by Chapman, 1913

Types of Slump



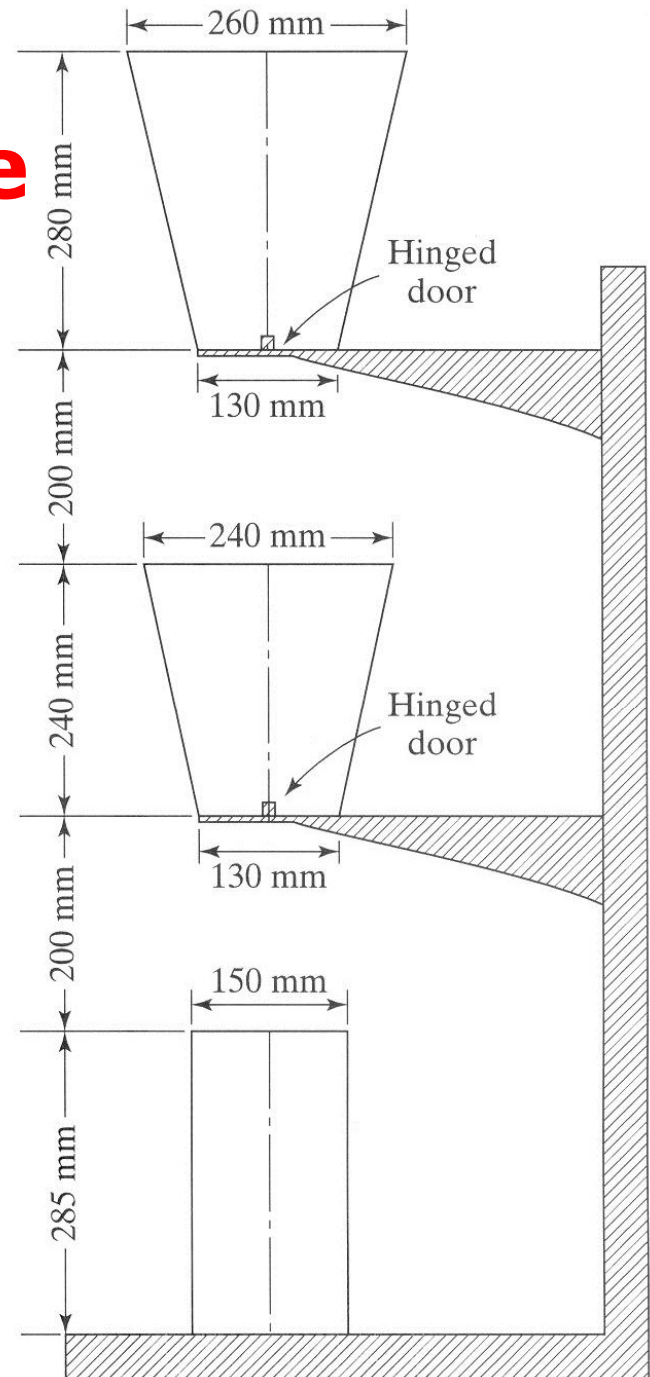
Compacting Factor Test: Principle

- The compacting factor test measures the **degree of compaction** resulting from the application of a **standard amount of work**, i.e. constant potential energy to kinetic energy



Compacting Factor Test: Apparatus and Procedure

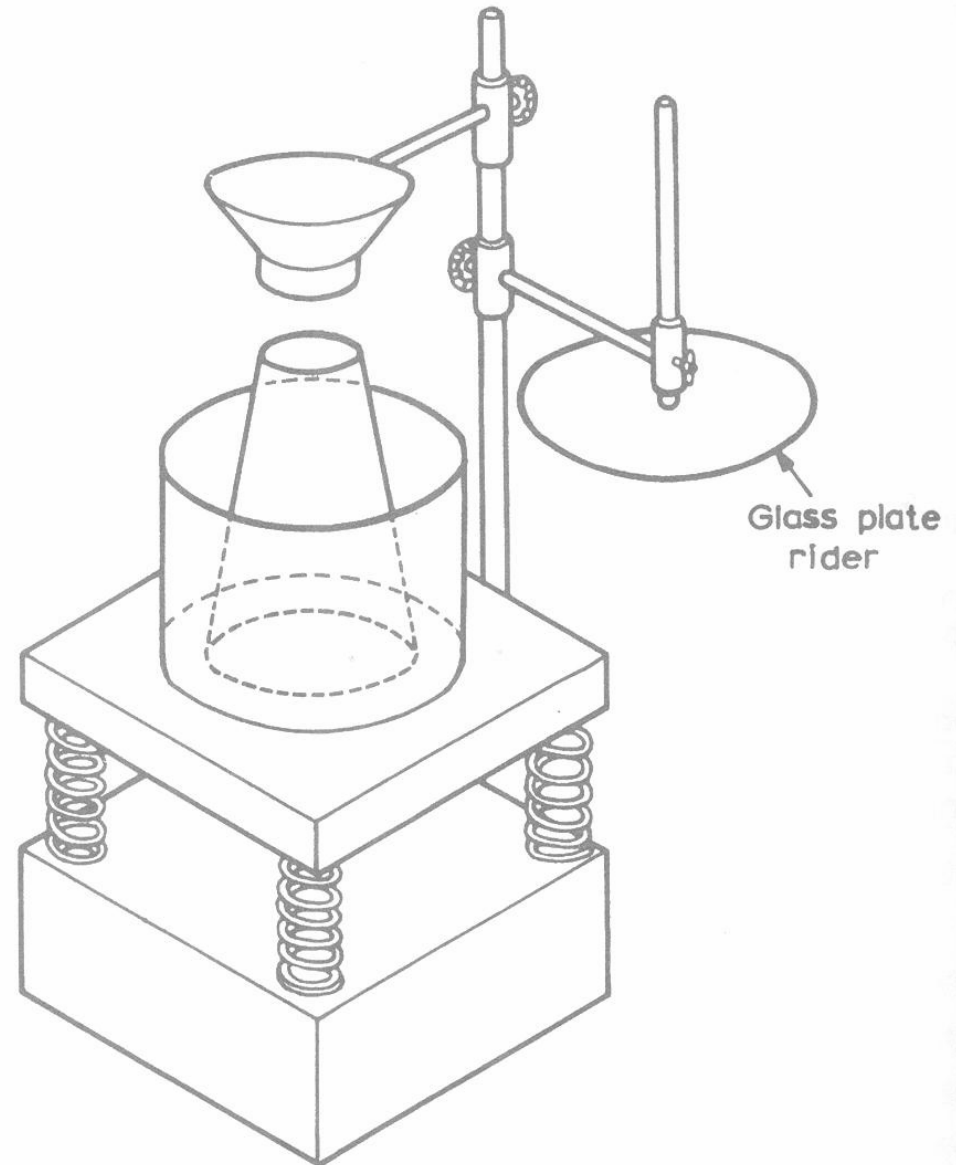
- Upper hopper is filled with concrete
- Bottom door of upper hopper is then released and concrete falls into lower hopper
- Bottom door of lower hopper is released and concrete falls into cylinder
- Excess concrete is cut and net weight of concrete in known volume of cylinder is determined
- The density of concrete in cylinder is now calculated, and this density divided by density of **fully compacted concrete** is defined as **Compacting Factor**



Vebe Test (ASTM C 1170): Principle



- The Vebe test measures the remolding ability of concrete under **vibration**
- It is assumed energy required for compaction is a measure of workability, and this is expressed in Vebe seconds, i.e. **time** required for remolding to be complete

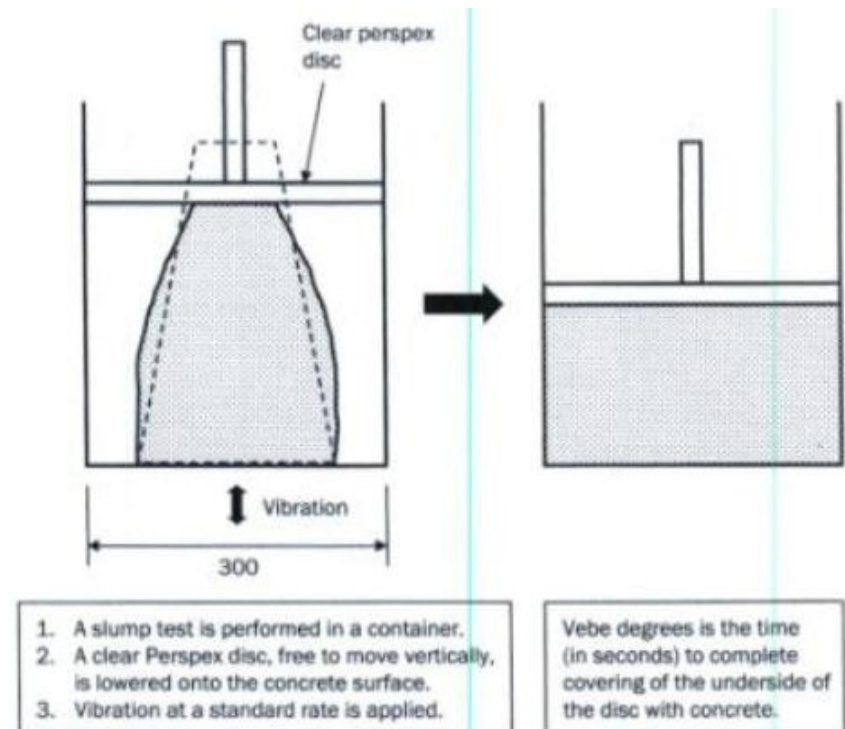


Vebe Test: Apparatus and Procedure



- A slump cone is placed and filled in the center of the cylinder
- After removing the slump cone, a glass plate is set atop the fresh concrete
- **Time** for the concrete to remold is recorded

<http://youtu.be/8n2wpCwYtq4>



The Vebe test.

Comparison of Workability Tests



- ❖ Slump test is suitable for concrete of **medium to high** workability
- ❖ Compacting factor test is more sensitive and accurate than slump test, especially for concrete mixes of **medium to low** workability
- ❖ Vebe test is a good laboratory test, particularly for concrete of **very low workability**

Suggested Values of Workability of Fresh Concrete for Different Placing Conditions



Placing condition	Degree of workability	Values of workability			
		Compacting factor, max. size of agg.			Vebe time, slump for 20 mm agg.
		10 mm	20 mm	40mm	
Hand compaction of heavily reinforced sections	High (flowing)	0.95	0.95	0.95	Vebe N/A 125-150 mm slump
Concreting of lightly reinforced section by hand or vibration of heavily reinforced sections	Medium (plastic)	0.88	0.90	0.92	5-2 s Vebe time, 25-75 mm slump
Concreting of lightly reinforced sections with vibration; road pavements and slabs with hand-operated vibrators and vibration of mass concrete	Low (stiff plastic)	0.82	0.84	0.85	10-5 s Vebe time, 5-50 mm slump
Concreting of shallow sections with vibrations	Very low (stiff)	0.75	0.78	0.80	20-10 s Vebe time, 0-25 mm slump
Concreting by intensive vibrations with vibropressing, centrifugation etc.	Extremely low (very stiff)	0.65	0.68	N/A	30-20 Vebe time Slump N/A



Thank you